



KTU NOTES

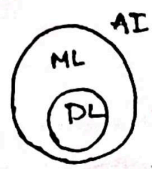
The learning companion.

**KTU STUDY MATERIALS | SYLLABUS | LIVE
NOTIFICATIONS | SOLVED QUESTION PAPERS**

 Website: www.ktunotes.in

Neural network } classifiers/ Regression.
SVM

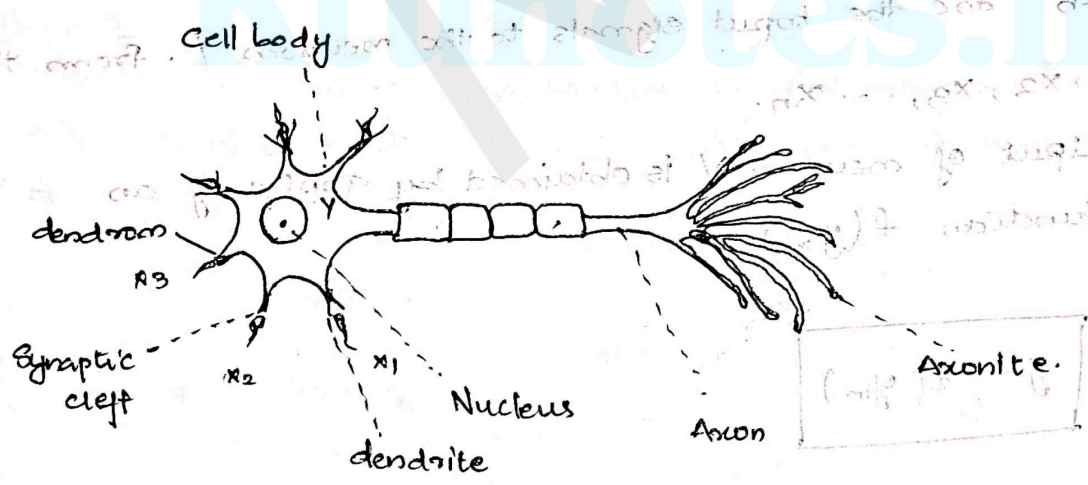
Artificial Neural network



ANN - Artificial neural network

- * An ANN is an information processing system that mimic the working of human brain
- * It consist of collection of artificial neurons [Mathematical model of neuron] arranged in layers

biological neuron



dendrites:- A dendrite receives input signal from other neurons and give it to the nucleus.

Nucleus:- It is the core part of cell body, which accepts input signals from dendrites and process it and produces a output pulse. If the i/p coming from dendrite is above a threshold value

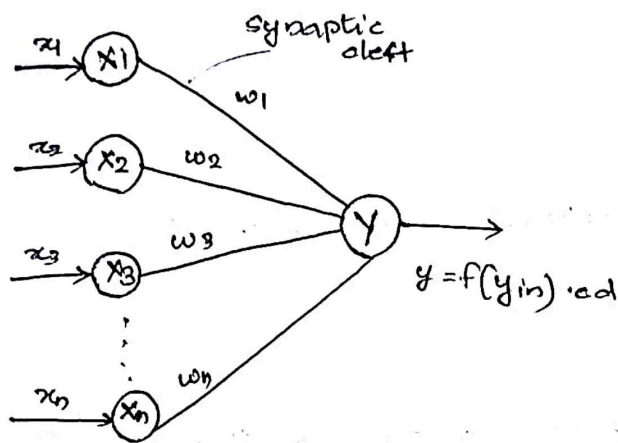
Then,

This pulse is send to the axon

Axon

An axon carries an output line. It carries information from the nucleus.

Architecture of / Mathematical model of a simple ANN



Here,

Net input to the neuron Y is y_{in}

$$y_{in} = x_1 w_1 + x_2 w_2 + \dots + x_n w_n$$

where,

$x_1, x_2, \dots, x_n$ are the input signals to the neuron Y from the neurons $x_1, x_2, x_3, \dots, x_n$.

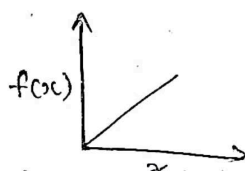
Then the output of neuron Y is obtained by applying an Activation function $f(y_{in})$

ie,

$$y = f(y_{in})$$

1. Identity function

$$f(x) = x$$



2) binary step function

$$f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{else} \end{cases}$$

3) Bipolar step function

$$f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ -1 & \text{else} \end{cases}$$

4. Sigmoidal function

2 types

* Binary: $f(x) = \frac{1}{1+e^{-x}}$

The range of binary sigmoidal function varies from

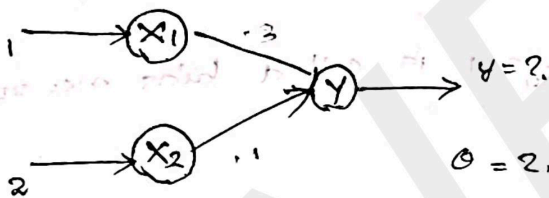
$$0 \rightarrow 1$$

* Bipolar: $f(x) = \frac{1-e^{-x}}{1+e^{-x}}$

The range is -1 to 1

It is used in back propagation network

a) Given a neural network



Find out the output of the neural network.

Assume,

Identity function is used ~~in the~~ as Activation function of the input neurons, and binary step function as activation function of output neurons.

$$\theta = 2$$

Input layer

AF = Identity

AF =

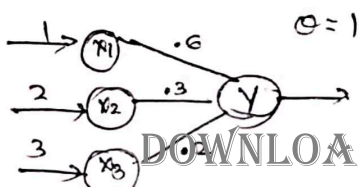
$$f(x) = \begin{cases} 1 & \text{if } x \geq \theta \\ 0 & \text{else} \end{cases}$$

$$y_{in} = 1 \times 0.3 + 2 \times 0.1 = 0.5$$

$$f(y) = \begin{cases} 1 & \text{if } 0.5 \geq 2 \\ 0 & \text{else} \end{cases}$$

$$\Rightarrow 0$$

a) Given a neural network



Find out the o/p of neuron.

Assume that identity function is used in the i/p layer
bipolar step function is used in the o/p layer with 0, 1

finds $y_{in} = 0.6 \times 1 + 2 \times 0.8 + 3 \times 0.2$

$= 1.8$

$$y = f(y_{in}) = \begin{cases} 1 & \text{if } 1.8 \geq 1 \\ 0 & \text{else} \end{cases}$$

Parameters

Threshold $\theta \rightarrow$

weight $\rightarrow$

AE $\rightarrow$

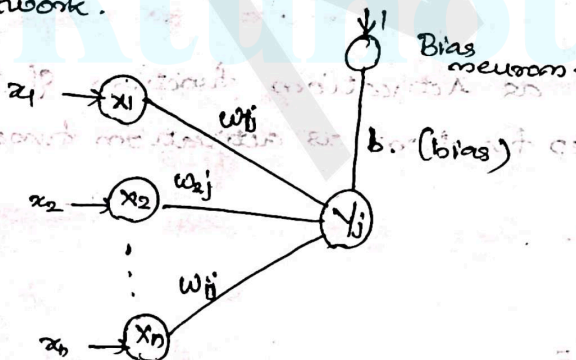
determine o/p.

Bias neuron and Bias

A neuron whose i/p is always 1 is called bias neuron

Applications

To implement the concept of line using an artificial neural network.



$$y_{in j} = 1 \times b + x_1 w_{1j} + x_2 w_{2j} + \dots + x_n w_{nj}$$

$$= b + \sum_{t=1}^n x_t w_{tj}$$

general equation for line $\Rightarrow y = mx + c$

This is an equation of line in the form.

$$y = mx + c$$

where,

y intercept is b

w $\rightarrow$ slope.

Terminologies in artificial neural network

1) weight

In an ANN neurons connected to other neurons through links or paths and a weight is attached with each link.

W_{ij} $\rightarrow$ represents the weight between i th neuron and the j th neuron

Threshold θ

it is ^{already} a set value [constant value] based upon which the final output of network is calculated.

For example:

$$f(y_{in}) = \begin{cases} 1 & \text{if } y \geq \theta \\ 0 & \end{cases}$$

weight

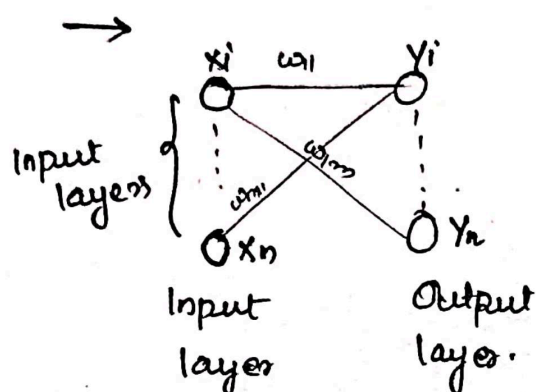
^{neural network}
An architecture learns to perform a task by adjusting
as changing ^{the} weight value

learning rate
weight α

It is used to control the amount of weight adjustment at each step while training a neural network

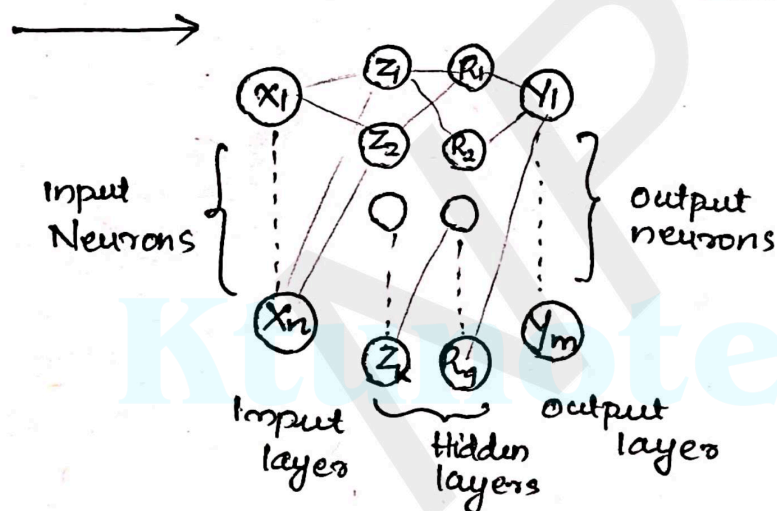
Wed 26/10/2022 Different ANN Architectures

(i) Single layer feed forward ANN



data flow
 $i/p \rightarrow o/p$
* Data flows in forward direction

(ii) Multilayer feed forward network



data flow
 $i/p \rightarrow o/p$
* data flows in forward directions

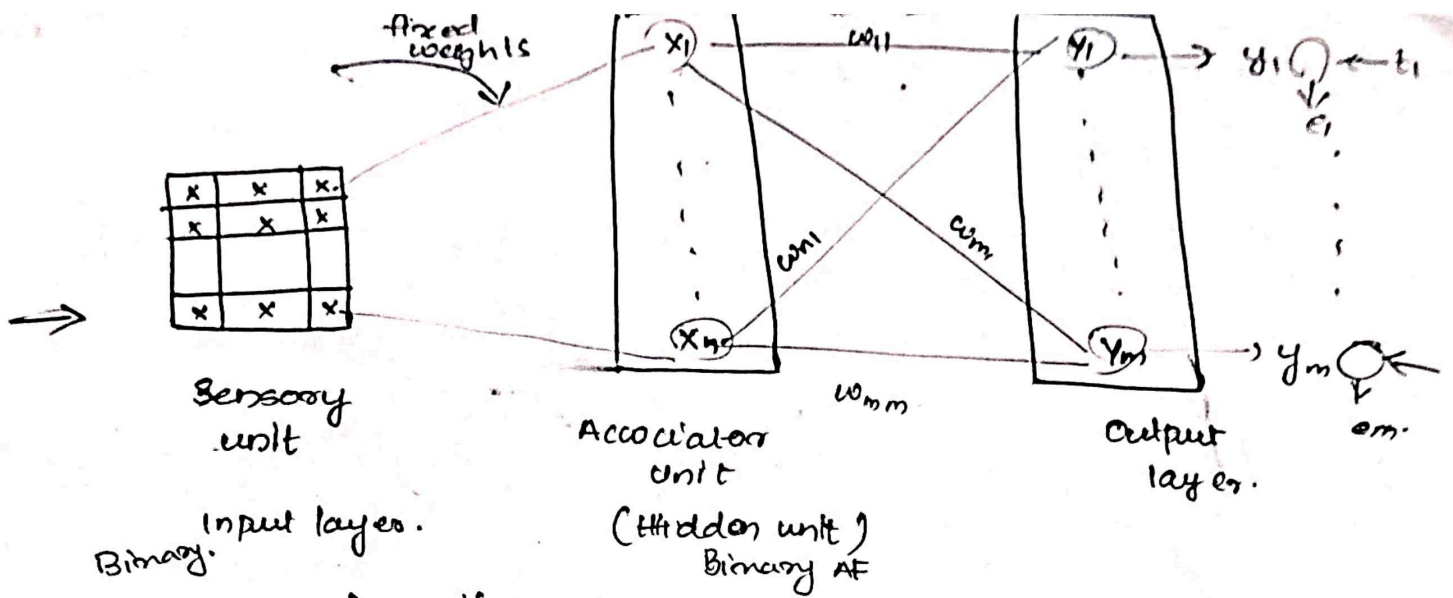
Supervised learning networks

Different supervised learning networks are

- ① Perceptron — [Single class
Multi class
- ② BPN (Back propagation Network)
- ③ ADALINE etc.

It is possible to train this N/w to perform a task
Each N/w uses its own training algorithm

Perceptron N/w



$$f(y_{im}) = \begin{cases} 1 & \text{if } y_{im} \geq 0 \\ 0 & \text{if } -\theta \leq y_{im} < 0 \\ -1 & \text{if } y_{im} < -\theta \end{cases}$$

Basic working

When an ip to the above n/w it calculate the o/p y . For calculating the final o/p. Below activation function is used.

Architecture Working

For each training ip the ~~final~~ network will calculate the final response [output]. and it will determine whether an error has occurred or not.

→ Error calculation is done by comparing the value of target t with the actual output y

→ If $y \neq t$ then the weights between associator unit should update below

$$w_i(\text{new}) =$$

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PHASE - II (Error calculation & back propagation)

Step 6: For each o/p unit y_k ($k = 1$ to m) compute error correction term δ_k .

$$\delta_k = (t_k - y_k) * f'(y_{in k}) = (t_k - y_k) * \underbrace{[f(y_{in k}) * 1 - f(y_{in k})]}_{\text{binary}}$$

on the basis of δ_k calculate change in weight $\Delta w_{jk} = \alpha \delta_k z_j$
 $\Delta w_{ok} = \alpha \delta_k$

Step 7: for each hidden unit z_j ($j = 1$ to p) calculate

$$\delta_{in j} = \sum_{k=1}^m \delta_k w_{jk}$$

$$\delta_j = \delta_{in j} * f'(z_{in j})$$

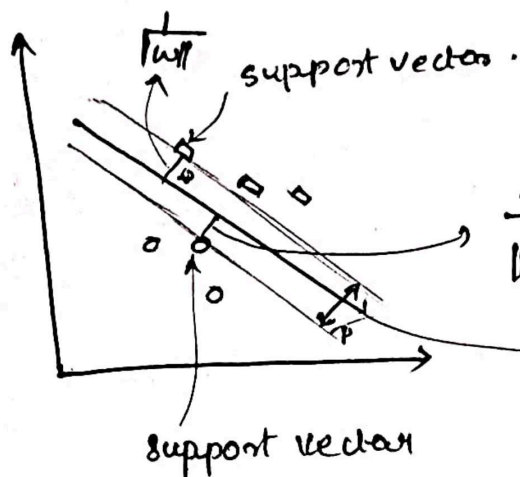
$$\Delta v_{ij} = \alpha \delta_j x_i$$

$$\Delta w_{oj} = \alpha \delta_j$$

PHASE III

01/11/2022
Tuesday
(Imp)

Hard Margin SVM



SVM for training data free of noise.

$$\text{Margin} = \frac{1}{\|w\|} + \frac{1}{\|w\|}$$

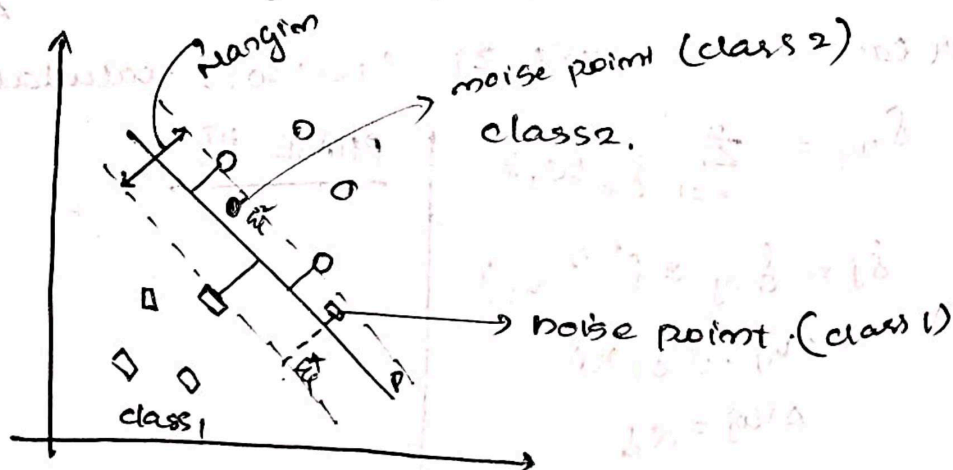
$$= \frac{2}{\|w\|}$$

In the case of an SVM the task is to find an optimal hyperplane subject to

- ① Maximize Margin.
ie. minimize $\frac{1}{2} \|w\|^2$
- ② $\gamma^t (w^t x^t + w_0) \geq \pm 1$

Soft margin hyperplane (Soft margin SVM)

* An SVM that can handle noise in the training data is called soft margin hyperplane



If the two classes are non-linearly separable because of the noise. Then instead of trying to find a non-linear SVM, we will try to find a SVM that incurs (causes) least errors. This type of SVM is called soft margin SVM.

We define a slack variable $\xi_j^t \geq 0$ which stores the deviation of a datapoint from the margin.

There are two types of deviation.

- i. An instance may lie on the wrong side of the hyperplane and is misclassified.
2. It may be on the right side but it lies in the margin.

So, the equation of the hyperplane that should be satisfied is

$$\gamma^t (w^T x^t + w_0) \geq 1 - \xi_j^t.$$

(i) If, $\xi_j^t = 0$

there is no problem with training data x^t

(ii) If, $0 < \xi_j^t < 1$

then x^t is correctly classified but it is in the margin

(iii) If, $\xi_j^t \geq 1$

then x^t is misclassified

We define a Soft error $\sum_t \xi_j^t$

Let, $L_p =$ penalty term

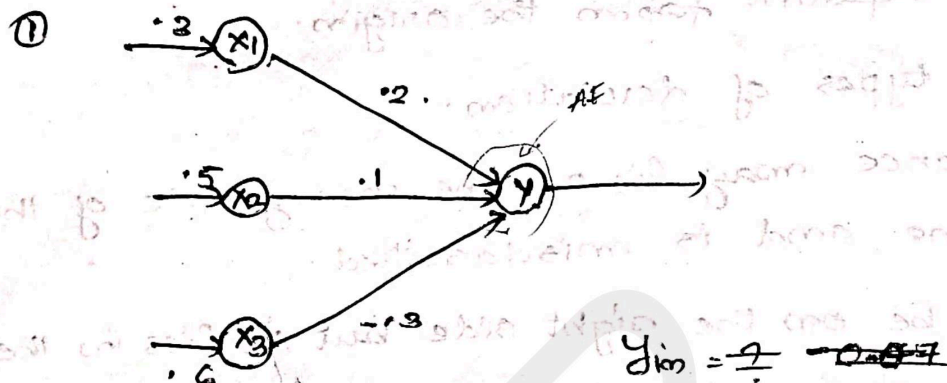
$$= C \sum_t \xi_j^t + \frac{1}{2} \|w\|^2$$

So, in the case of a soft margin SVM the objective is to minimize L_p subject to find a line $\gamma^t (w^T x^t + w_0) \geq 1 - \xi_j^t$

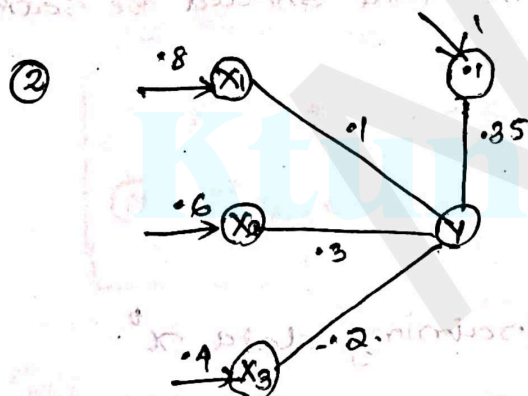
Objective!

$$\text{minimize} = \frac{1}{2} \|w\|^2 + \sum_i \xi_i^2$$

Q) Calculate net i/p to the neuron.



$$= 0.06 - 0.18 = -0.12$$



① Suppose $AF = \begin{cases} 1 & \text{if } y_{in} \leq 0 \\ 0 & \text{else} \end{cases}$

$$\theta = 1$$

then o/p of neuron y ?

$$0.7 \leq 1$$

$$y_{in} \leq 0$$

$$= 1$$

Q) calculate the net i/p to the neuron Y and use binary sigmoidal activation function.

AF = binary sigmoidal.

$$f(y_{in}) = \frac{1}{1 + e^{-y_{in}}}$$

$$y_{in} = 1 \times 0.35 + 0.8 \times 0.1 + 0.6 \times 0.3 + 0.4 \times 0.2 = 0.53$$

$$y = f(y_{in}) = \frac{1}{1 + e^{-0.53}} = 0.625$$

Use bipolar sigmoidal function.

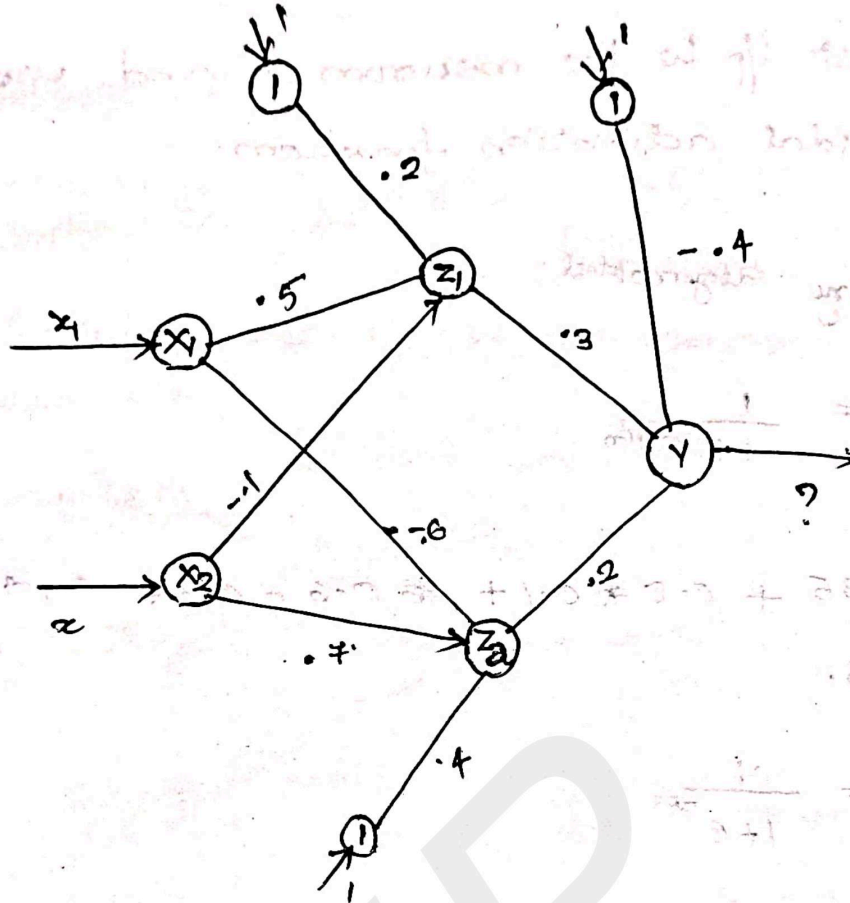
bipolar sigmoidal

$$f(y_{in}) = \frac{2}{1 + e^{-y_{in}}} - 1$$

$$= \frac{2}{1 + e^{-0.53}} - 1$$

$$= 0.259$$

BPN



(i) Given an i/p $[1 \ 0]$ & target o/p 1

(ii) Find the o/p of feed forward phase.

Assume that,

The identity function is used as the activation function of i/p layer. So the o/p of neurons x_1 and x_2 $x_1 = 1$ & $x_2 = 0$

(*) Net i/p to neurons z_1 & z_2

$$Z_{in1} = 1 \times 0.2 + 1 \times 0.5 + 0 \times 0.1 = \underline{\underline{0.7}}$$

$$Z_{in2} = 1 \times 0.4 + 0 \times 0.7 + 1 \times 0.6 = \underline{\underline{-0.2}}$$

(*) $z_1 = f(z_{in1})$

$$= \frac{1}{1 + e^{-z_{in1}}}$$

$$= \frac{1}{1 + e^{-0.7}}$$

$$= \underline{\underline{0.33}}$$

$$z_2 = f(z_{n2})$$

$$= \frac{1}{1 + e^{0.2}}$$

$$= \underline{\underline{0.55}}$$

⊛ Net i/p to neuron y

$$y_{in} = 1 * -0.4 + z_1 * 0.3 + z_2 * 0.2$$

$$= 1 * -0.4 + 0.33 * 0.3 + 0.55 * 0.2$$

$$= \underline{\underline{-0.19}}$$

⊛ $y = f(y_{in})$

$$= \frac{1}{1 + e^{-y_{in}}}$$

$$= \frac{1}{1 + e^{+0.19}}$$